Shubham Verma

Lead Data Scientist · Applied ML & GenAI Systems · Advisor, AI/DS Chapter

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SUMMARY

ML and GenAI engineering lead with 6+ years building production AI for industrial asset monitoring. I own systems end to end, from problem framing and architecture through distributed engineering, deployment and team delivery. At Mechademy I designed and led the platforms that train, evaluate, deploy and retire thousands of equipment health models automatically, cutting model turnaround from weeks to hours. After leading a team of 8, I now work as a senior individual contributor and advise data science teams across the company through the AI/DS Chapter. In 2026 I single-handedly built the company's production agent platform, shipping 13 LLM agents in five weeks.

FLAGSHIP SYSTEMS · DESIGN AND ENGINEERING LEAD

Agent Lifecycle Loop · production platform running a fleet of LLM agents for industrial operators — May 2026 – present · architect & sole lead engineer

- Designed and built the platform from scratch: a composition runtime, an agent fleet, a capability-governance layer and a control plane, joined in one declare → test → publish → run → observe → judge loop.
- Shipped **13 production agents in five weeks**, 8 delivering branded PDF reports, from model-health triage to diagnostic-rule review and per-customer value reports.
- Made durability, tracing, cost metering and access control come from configuration instead of agent code: every agent step retries on its own, every token has an owner and a price, and each agent reaches only the tools it is granted.
- Built a privacy airlock that de-identifies customer data before it reaches any model or stored state; moved its NER model to ONNX Runtime with a custom decoder: **5× faster, ~50 MB instead of 1 GB**, identical output.
- Turned every production failure into a rule, not a patch: 64 architecture decision records, 3,400+ offline tests, and a cookbook whose examples execute in CI.

Python · LangGraph · Temporal · LiteLLM · Langfuse · MCP · Django · Next.js · Postgres · GLiNER/Presidio on ONNX Runtime · Helm/EKS

ML Lifecycle Loop · self-running platform that creates, validates, deploys, monitors and retrains a fleet of industrial time-series models

- Turned model creation from an engineer-bound problem into a compute-bound one: **~$0.08 of compute per model**, with headcount no longer scaling with model count.
- Encoded domain expertise as SME-validated feature definitions plus a feature store, so onboarding a new client became a lookup instead of a fresh derivation.
- Made operating-regime selection a first-class step, the single biggest accuracy lever: **roughly 30–50% better than generic AutoML**.
- Built a crash-resilient job bridge onto a distributed Ray cluster, so multi-hour training runs survive head-node failures.
- Designed a drift backbone using four statistical tests in a layered decision: ignore transient blips, correct systematic bias without retraining, retrain only on real degradation.

Python · Ray (KubeRay) · Dagster · MLflow · XGBoost · LightGBM · PyTorch · TimescaleDB · Iceberg on S3 · Gemini

Evaluation-Driven AutoML · AutoML where evaluation decides which features survive and which models ship

- Built a dual-gate feature selector separating training-time gain from cross-validated generalization, cutting features that look strong but fail on held-out data.
- Added multi-pass LLM feature selection that brings physical domain judgment while reasoning only over masked, generic feature types, never client or facility identifiers.
- Introduced a vision-language model as judge: every trained model is reviewed against recent live data and routed to deploy, retrain or escalate, catching models that pass numeric metrics but visibly mistrack.

Python · Gemini · XGBoost · LightGBM · scikit-learn · Ray · MLflow

Curve Digitizer · production computer-vision pipeline turning scanned performance curves into calibrated numeric data

- Unlocked engineering data that existed only as scanned graphs, removing a manual digitisation step that blocked downstream models and simulations.

- Orchestrated foundation models per sub-problem (a vision-LLM calibrates axes and units, SAM lifts curves) with a 10-stage signal-processing chain and an **$R^2 \geq 0.95$** acceptance gate.
- Solved intersecting curves, where off-the-shelf digitisers fail, with direction-aware separation and a fast human-in-the-loop fallback.

Python · SAM · Gemini · OpenCV · FastAPI · AWS EKS

EXPERIENCE

Mechademy · Industrial AI for turbomachinery and rotating-equipment health — New Delhi · Feb 2020 – present

Lead Data Scientist & Advisor, AI/DS Chapter — Current

- Promoted from team lead to a senior individual-contributor role, owning architecture of the flagship ML and GenAI platforms above.
- Advise data science teams across the company through the AI/DS Chapter on modelling standards, evaluation practice and platform adoption.

Data Science Team Lead

- Led a team of 8 data scientists and ML engineers; set technical direction, reviewed architecture and owned delivery of the company's core ML platforms.
- Designed a GenAI diagnostics system that drafts fault rules, consolidates fault knowledge and generates root-cause analyses for industrial components (LangGraph, Pydantic, Langfuse).
- Built a fault-to-sensor mapping layer linking sensor behaviour patterns to diagnostic tests, giving engineers traceable, explainable alerts.
- Created a rapid-prototyping framework that standardises how GenAI pipelines are written, validated, logged and evaluated.
- Designed adaptive anomaly-detection methods, including real-time dynamic thresholds and self-tuning outlier detection, reducing manual tuning and false alarms.

Senior Machine Learning Engineer

- Built the first AutoML proof of concept on cloud processing jobs, the foundation of the ML Lifecycle Loop.
- Owned batch ETL and ML inference pipelines feeding live equipment monitoring.
- Developed deep-learning health monitors (image-based autoencoders, multivariate reconstruction models) and an event engine that groups raw alerts into actionable events.

Machine Learning Engineer — From Feb 2020

- Delivered tabular ML models and EDA insights across many equipment use cases; introduced MLflow-based experiment tracking to the team.

SKILLS

ML & GenAI: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, AutoML, anomaly detection, time-series modelling, LangGraph, LangChain, LiteLLM, MCP, Langfuse, LLM/VLM evaluation, SAM, OpenCV

Platforms & MLOps: Ray/KubeRay, Temporal, Dagster, Dask, MLflow, Celery, Docker, Kubernetes, Helm, AWS (EKS, S3, SageMaker), ONNX Runtime

Data & web: Python, SQL, Postgres, TimescaleDB, Iceberg, DuckDB, MongoDB, Pandas, Django, FastAPI, Next.js

Leadership: system architecture, technical roadmapping, architecture decision records, mentoring, cross-functional delivery with domain engineers

EDUCATION

B.Tech, Mechanical Engineering · Maharshi Dayanand University, Rohtak — 2014 – 2018